## Multi-dimensional Array

A multi-dimensional array is an array with more than one dimension. For example, a 2D array is an array of arrays, meaning it is a matrix of rows and columns. A 3D array adds another dimension, turning it into an array of arrays of arrays.

## 2 Dimensional Array

A **two-dimensional array**, also known as **2D array**, is a collection of elements arranged in a matrix with rows and columns.

**Example program to display the elements in a 2D array**

```
void main( )
{
        int arr[2][3] = { {10, 20, 30},
                          {40, 50, 60} };
        int i, j;
        for (i = 0; i < 2; i++)
        {
            for (j = 0; j < 3; j++)
            {
                printf("%d ", arr[i][j]);
            }
            printf("\n");
        }
}
```

Output

```
10  20  30
40  50  60
```

**Example program to take the elements of a 2D array as user input and display them to the output screen**

```
void main( )
{
        int arr[10][10];
        int i,j,m,n;
        printf("\nEnter the number of rows ");
        scanf("%d",&m);
        printf("\nEnter the number of columns ");
        scanf("%d",&n);
        printf("Enter the elements of the array ");
        for (i = 0; i < m; i++)
        {
             for (j = 0; j < n; j++)
             {
                  scanf("%d", &arr[i][j]);
             }
        }
        printf("\n The elements of the array are \n ");
        for (i = 0; i < m; i++)
        {
             for (j = 0; j < n; j++)
             {
                  printf("%d  ", arr[i][j]);
```

```
            }
             printf("\n");
        }
}
```

**C Program to add 2 matrices**

```
void main( )
{
        int a[20][20], b[20][20], sum[20][20];
        int i,j,r,c;
        printf("\nEnter the number of rows  ");
        scanf("%d", &r);
        printf("\nEnter the number of columns   ");
        scanf("%d", &c);
        printf("\nEnter the elements of 1st matrix:\n");
        for (i = 0; i < r; i++)
        {
            for (j = 0; j < c; j++)
            {
                scanf("%d", &a[i][j]);
            }
        }
        printf("Enter the elements of 2nd matrix:\n");
        for (i = 0; i < r; i++)
```

```
        {
                for (j = 0; j < c; j++)
                {
                        scanf("%d", &b[i][j]);
                }
        }
    for (i = 0; i < r; i++)
    {
        for (j = 0; j < c; j++)
        {
                sum[i][j] = a[i][j] + b[i][j];
        }
    }
     printf("\nSum of two matrices is: \n");
     for (i = 0; i < r; i++)
    {
            for (j = 0; j < c; j++)
           {
                        printf("%d", sum[i][j]);
           }
        printf("\n");
    }
}
```

Output

Enter the number of rows  2

Enter the number of columns  3

Enter the elements of 1st matrix:

1 2 3 4 5 6

Enter the elements of 2nd matrix:

10 20 30 40 50 60

Sum of two matrices is:

11  22  33

44  55  66

**C Program to find the transpose of a matrix**

```
void main( )
{
        int a[20][20], t[20][20];
        int i,j,r,c;
        printf("\nEnter the number of rows  ");
        scanf("%d", &r);
        printf("\nEnter the number of columns   ");
        scanf("%d", &c);
        printf("\nEnter the elements of the matrix:\n");
        for (i = 0; i < r; i++)
        {
                for (j = 0; j < c; j++)
```

```
            {
                scanf("%d", &a[i][j]);
            }
        }
        printf("The elements of the matrix are:\n");
        for (i = 0; i < r; i++)
        {
            for (j = 0; j < c; j++)
            {
                printf("%d ", a[i][j]);
            }
            printf("\n");
        }
    for (i = 0; i < r; i++)
    {
        for (j = 0; j < c; j++)
        {
            t[j][i] = a[i][j];
        }
    }
    printf("\nThe transpose matrix is: \n");
    for (i = 0; i < c; i++)
    {
```

```
            for (j = 0; j < r; j++)
            {
                        printf("%d", t[i][j]);
            }
        printf("\n");
    }
}
```

<u>Output</u>

Enter the number of rows  3

Enter the number of columns  2

Enter the elements of the matrix:

1 2 3 4 5 6

The elements of the matrix are:

1 2

3 4

5 6

The transpose matrix is:

1  3  5

2  4  6

## C Program to multiply 2 matrices

```
#include<stdio.h>
void main( )
{
	int a[10][10],b[10][10],mult[10][10];
	int i,j,k,r,c;
	printf("enter the number of rows\n");

	scanf("%d",&r);
	printf("enter the number of columns\n");
	scanf("%d",&c);
	printf("enter the elements of first matrix\n");
	for(i=0;i<r;i++)
	{
		for(j=0;j<c;j++)
		{
			scanf("%d",&a[i][j]);
		}
	}
	printf("enter the elements of second matrix\n");
	for(i=0;i<r;i++)
	{
		for(j=0;j<c;j++)
		{
			scanf("%d",&b[i][j]);
		}
	}
```

```
        for(i=0;i<r;i++)
        {
                for(j=0;j<c;j++)
                {
                        mult[i][j]=0;

                        for(k=0;k<c;k++)
                        {
                                mult[i][j]=mult[i][j]+a[i][k]*b[k][j];
                        }
                }
        }
        printf("multiplication of the matrix is\n");

        for(i=0;i<r;i++)
        {
                for(j=0;j<c;j++)
                {
                        printf("%d\t",mult[i][j]);
                }
                printf("\n");
        }
}
```